

“E-commerce customer dataset”

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Introduction

Project Background

This project focuses on analyzing customer data to better understand customer behavior, preferences, and purchasing patterns. The dataset contains information about customers' demographics, membership tiers, purchasing activity, spending, satisfaction, and interaction with different sales factors such as discounts, devices, and payment methods.

The main goal of the analysis is to turn the raw customer data into useful business insights through data visualization. The analysis explores customer profiles, identifies locations with stronger business potential, finds customers who may be suitable for loyalty programs, detects valuable customers who are at risk of becoming inactive, and examines how different factors may influence customer behavior.

The visualizations were designed to present these insights in a clear and accessible way, helping decision-makers quickly understand customer trends and identify areas where targeted business actions could improve customer retention, engagement, and overall revenue.

Business Objectives

The analysis aims to answer the following business questions:

- Who are the company's most valuable customers?
- Which customers are at risk of churn?
- Which cities are the best candidates for future marketing campaigns?
- Are discount campaigns actually effective?
- What practical actions can increase future sales?

Dataset Description

Column	Data Type	Description
customer_id	Integer	Unique identifier assigned to each customer.
first_name	String	Customer's first name.
gender	Category	Customer's gender (M = Male, F = Female).
age	Integer	Customer's age in years.
city	Category	City where the customer resides.
province	Category	Province where the customer resides.
signup_date	Date	Date when the customer registered.
membership_tier	Category	Customer's membership level (e.g., Silver, Gold, VIP).
purchase_count	Integer	Total number of purchases made by the customer.
avg_order_value	Float	Average monetary value of each purchase.
total_spending	Float	Total amount spent by the customer across all purchases.
last_purchase_days	Integer	Number of days since the customer's most recent purchase.
payment_method	Category	Customer's preferred payment method.
device	Category	Device used to make purchases (e.g., Android, iPhone, Web).
discount_used	Category	Indicates whether the customer used a discount (Yes or No).

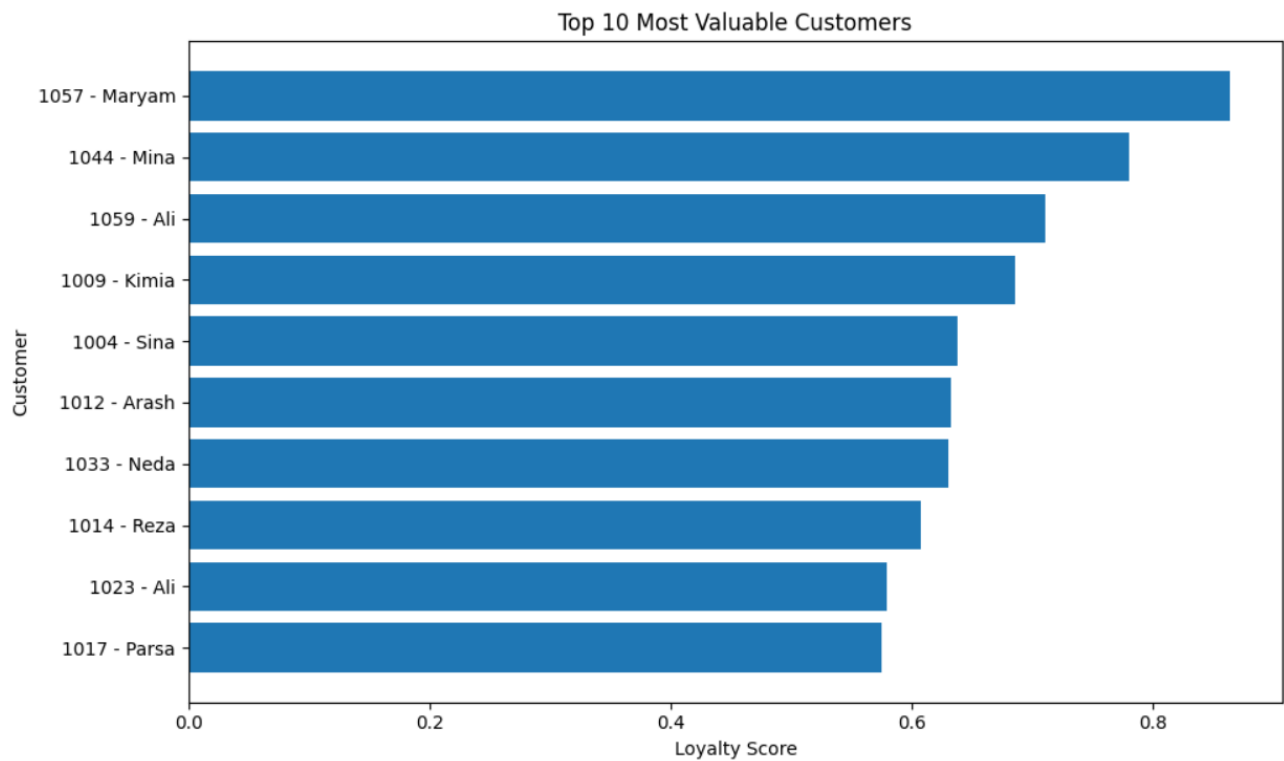
returned_items	Integer	Number of items returned by the customer.
satisfaction_score	Integer	Customer satisfaction rating on a scale from 1 (lowest) to 5 (highest).

Business Analysis

Who are the most valuable customers?

Analysis Approach

- Analyzed total spending, purchase count, recency, and satisfaction.
- Normalized these factors and combined them into a Loyalty Score.
- Selected the top 10 customers with the highest scores.
- Used a scatter plot to compare purchase count and total spending.
- Point size represents customer activity/recency.



Observation

- The most valuable customers showed higher spending and more frequent purchases.
- Customers with recent purchases were given more importance in the loyalty score.
- The analysis identified 10 customers as the main candidates for loyalty programs.

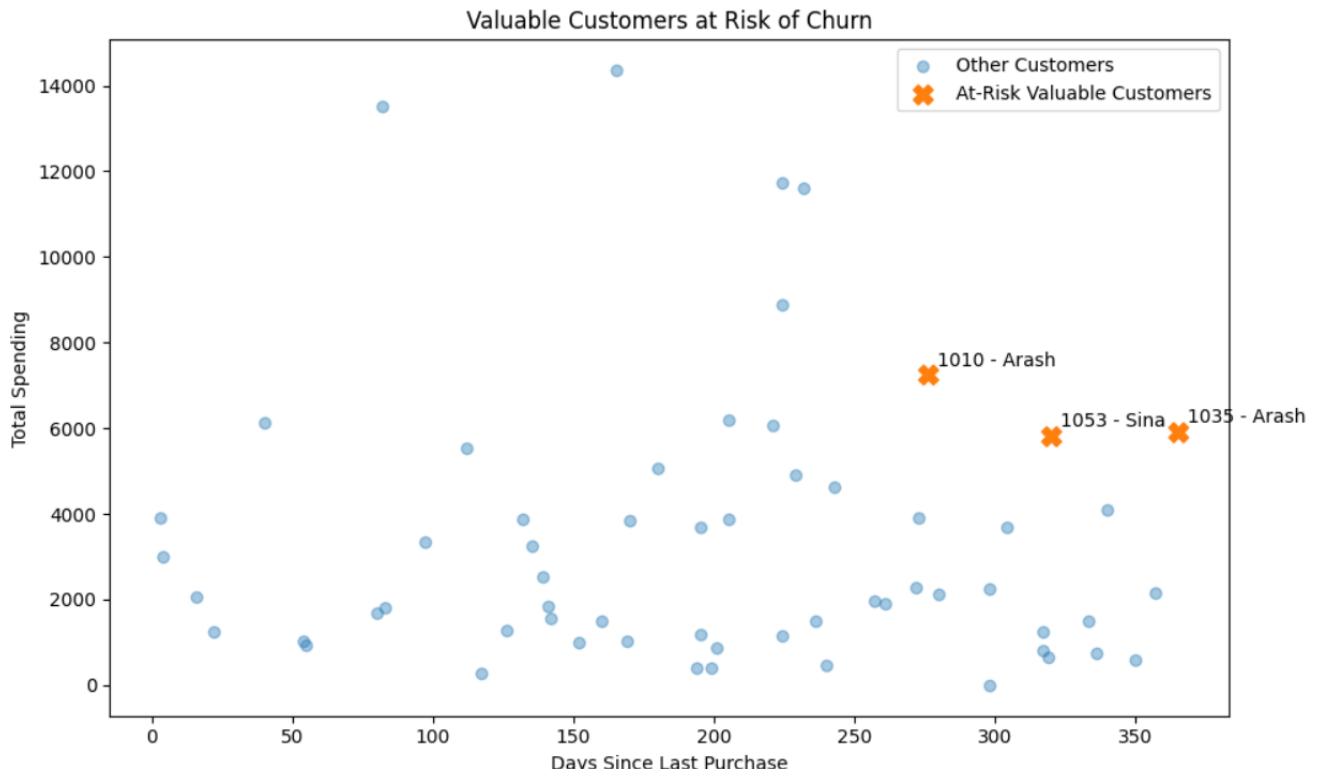
Business Impact

Identifying valuable customers helps the business focus its loyalty efforts on customers who are most likely to generate continued revenue. These customers can be targeted with personalized rewards, exclusive offers, or retention campaigns, helping to increase customer retention, encourage repeat purchases, and maintain long-term revenue.

Which customers are at the risk of churning?

Analysis Approach

- Identified high-value and frequent customers based on spending and purchase count.
- Used days since last purchase to measure inactivity.
- Customers meeting all three conditions were classified as at-risk.
- Used a scatter plot to compare customer spending and inactivity.



Observation

- A number of high-value customers had not purchased recently.
- These customers were identified as potential churn risks.
- The chart highlights which customers should receive immediate attention.

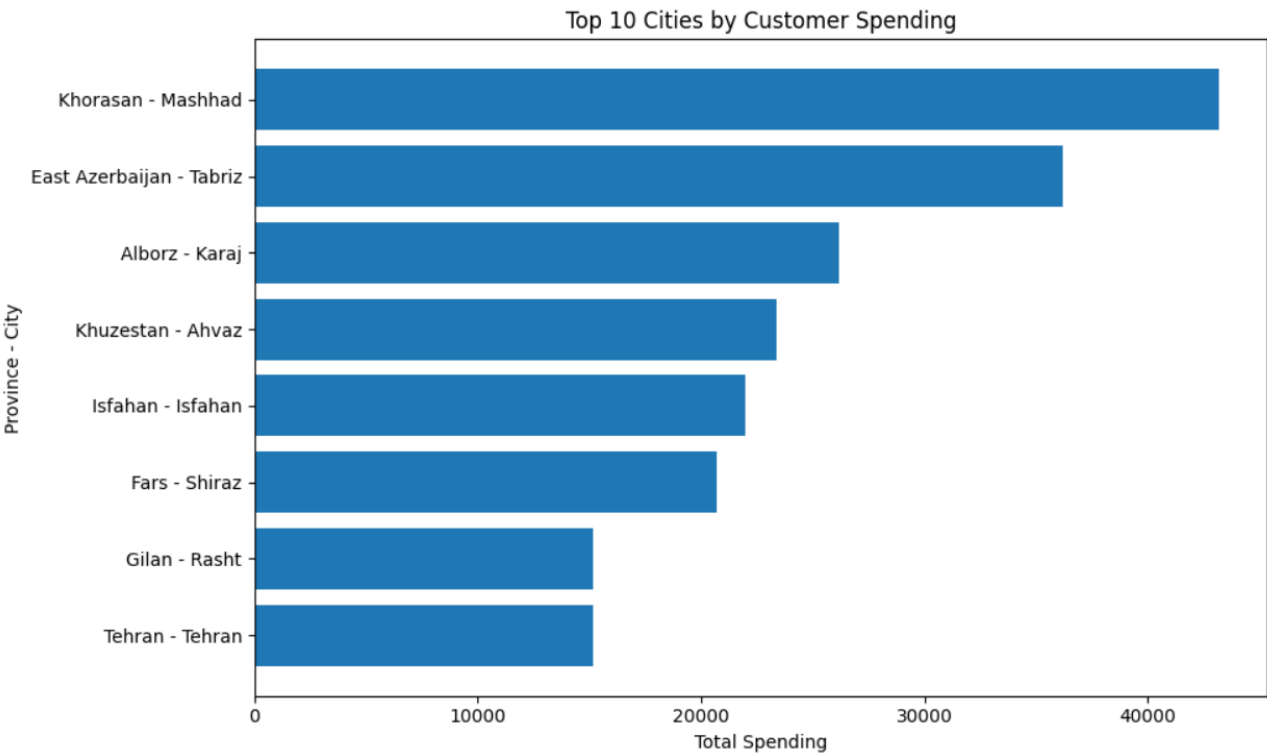
Business Impact

- Helps the business identify valuable customers before they become inactive.
- Enables targeted retention campaigns and personalized offers.
- Can help protect existing revenue and improve customer retention.

Which cities are best for marketing?

Analysis Approach

- Grouped customers by province and city.
- Calculated total customer spending and customer count for each city.
- Focused on cities with at least 3 customers.
- Identified the top 10 cities by total spending.
- Compared city spending with average customer satisfaction.



Observation

- The top-performing cities generated the highest total customer spending.
- Cities with more customers generally had greater total spending.
- The analysis highlights the cities with the strongest potential for targeted marketing.

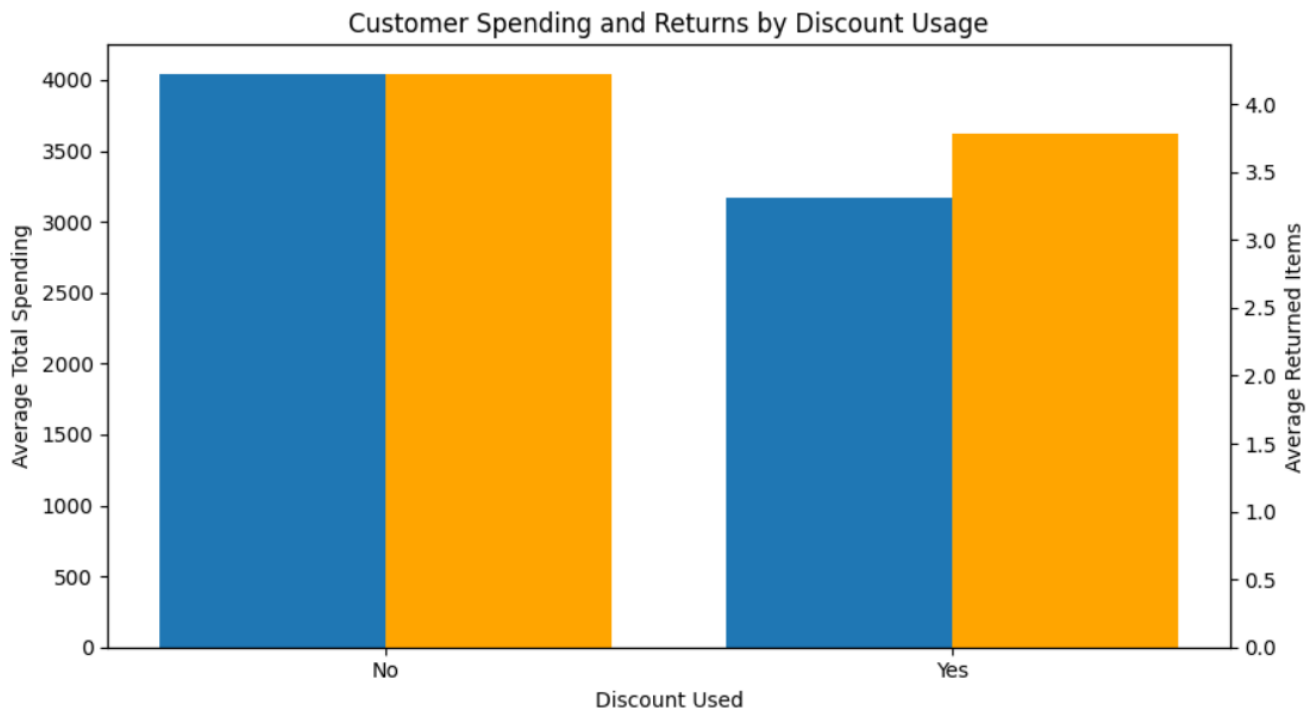
Business Impact

- Helps the business prioritize marketing budgets in high-performing cities.
- Can increase the potential return on advertising investment.
- Provides a data-driven basis for selecting locations for future marketing campaigns.

Are discount really effective?

Analysis approach

- Compared customers who used discounts with those who did not.
- Calculated the average total spending and average returned items for both groups.
- Used a dual-axis bar chart to compare the two metrics despite their different scales.



Observation

- Customers who used discounts can be compared directly with customers who did not.
- The chart shows whether discount users have higher spending and whether this is accompanied by more returned items.
- The results indicate whether the increase in spending is worth the potential increase in returns.

Business Impact

- Helps determine whether discounts are effective in increasing customer spending.
- Helps the business balance higher sales against product returns.
- Supports better decisions about when and how discounts should be offered.

Business recommendation

Focus marketing on high-performing cities

Evidence:

The analysis identified cities with the highest total customer spending, showing that some locations generate significantly more revenue than others.

Action:

Allocate a larger share of the marketing budget to the highest-performing cities and run targeted campaigns based on their customer profiles.

KPI:

Measure revenue generated per city and marketing ROI before and after the campaigns.

Build a loyalty program for high-value customers

Evidence:

The loyalty analysis identified customers with high spending, frequent purchases, recent activity, and higher satisfaction as the most valuable customer segment.

Action:

Offer these customers personalized rewards, exclusive offers, or points based on purchase frequency and spending to encourage repeat purchases.

KPI:

Track repeat purchase rate, average customer spending, and customer retention rate.

Use discounts selectively

Evidence:

The discount analysis showed differences in average spending and returned items between customers who used discounts and those who did not. This allows the company to evaluate whether higher spending justifies potential additional returns.

Action:

Offer discounts selectively to customer segments where they generate additional spending, rather than applying discounts broadly to all customers.

KPI:

Track incremental revenue, average order value, and return rate for discounted purchases.

Tools used

- Jupyter notebook
 - Python
 - Pandas
 - Matplotlib
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Conclusion

This project used customer data to identify purchasing patterns, valuable customers, high-performing locations, and factors affecting customer behavior. The analysis showed that customer spending, purchase frequency, recency, satisfaction, and location can provide useful insights for business decisions.

The results can help the company focus its marketing efforts on high-performing cities, build loyalty strategies for valuable customers, and use discounts more selectively. Overall, the analysis demonstrates how customer data can be transformed into practical insights to support sales growth, customer retention, and better marketing decisions.